805. “Grete Hermann pointed out a glaring deficiency in the argument, but she seems to have been entirely ignored. Everybody continued to cite the von Neumann proof.” For more on Hermann, see the presentation slides by M. P. Seevinck on Grete Hermann (2012). Also see the resources at http://web.mit.edu/redingtn/www/netadv/PHghermann.html, accessed September 20, 2017.

hole at its heart: See Jammer 1974, p. 247: “In spite of the opposition to Bohr’s views by some leading physicists like Einstein and Schrödinger the vast majority of physicists accepted the complementarity [i.e. Copenhagen] interpretation in general without reservations, at least during the first two decades after its inception.”

a number for every point in space: For specialists: I’m just using position-space wave functions for single-particle stationary states as an example. I’ll get to more complex stuff later.

it’d display 0.02: Sometimes the Wave-Function-O-Meter™ might display an imaginary number, like the square root of negative one. But put that complication aside for now.

the *probability* that the electron is in that place: Technically, it’s the square of the wave function that gives you the probability, but the idea is the same.

“does that change the [quantum] state of the universe?”: Walter Isaacson 2007, *Einstein: His Life and Universe* (Simon and Schuster), p. 515.

“highly qualified measurer—with a Ph.D.?”: Bell 2004, p. 117.

Chapter 2

“at greater length”: Heisenberg 1971, p. 62.

came to be known as “quantum physics”: Bohr’s model of the atom was not the origin of the term “quantum physics.” The term slowly came into use over the first decade of the twentieth century, as various phenomena were discovered that involved the absorption or emission of discrete packets of electromagnetic radiation, starting with Planck’s law of black-body radiation. This period in the history of physics that I’m quickly breezing through—from Planck’s discovery in 1900 to the theories that Heisenberg and Schrödinger developed in 1925, which are described in this chapter—is worthy of a book in its own right. Many have been